them to the PMO. Once the PMO accepts the panel's suggestions, the MoD and DRDO have a 90-day time-frame to start implementing them. Corridors of the DRDO headquarters in New Delhi are abuzz with views on how to meet the tough deadline, as many scientists believe that it is not easy to shake up a 66-year-old institution in months.

Besides VijayRaghavan, the committee included Lt Gen. Subrata Saha (retd), former deputy chief of army staff; Vice Admiral S.N. Ghormade, former vice chief of navy staff; Air Marshal B.R. Krishna, former chief of integrated staff; Sujan R. Chinoy, director general of the Manohar Parikkar Institute of Defence Studies and Analyses; Prof. Manindra Agarwal of IIT Kanpur; S.P. Shukla, president, Society of Indian Defence Manufacturers; J.D. Patil of L&T Defence; S. Unnikrishnan Nair, distinguished scientist, ISRO, and Rasika Chaube, financial advisor in the MoD.

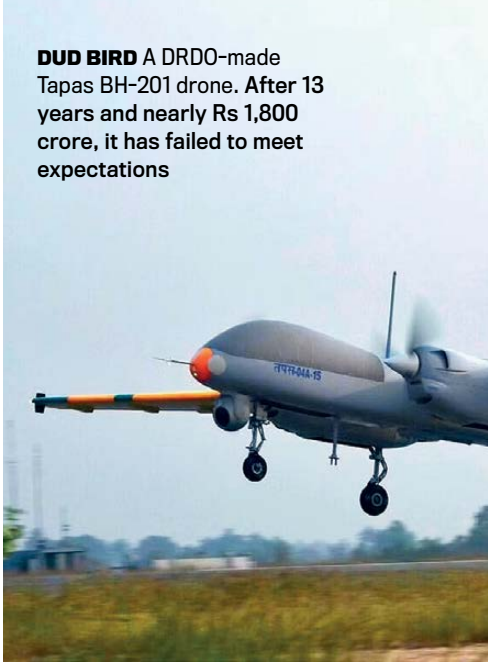
Ravi Gupta, a former DRDO scientist, is sceptical about the panel's suggestions. "The committees formed to restructure DRDO were headed by outsiders who have no idea about defence research and DRDO's functioning. The committees made suggestions without gauging their repercussions. It takes decades to build an organisation but does not take long to destroy it," says Gupta. The DRDO has a budget of Rs 23,264 crore in the budget estimate (BE) 2023-24, with nearly 900 projects at various stages of development.

## WHAT THE COMMITTEE RECOMMENDS

The VijayRaghavan panel has advocated a pivotal role for the Defence Technology Council (DTC), which is headed by the prime minister and has the defence minister and the national security advisor as members, to identify suitable players for specific defence technologies. The inclusion of two members each from academia and industry in the DTC is also recommended to bring diverse perspectives. Not only does this mark the PMO's entry in defence research and direct involvement in strategically important projects, but also oversight over DRDO. The panel also suggests a separate department under the MoD—the department of defence science, technology and innovation (DDSTI)—to promote defence R&D in academic and start-up ecosystems. It will also serve as the DTC secretariat. In addition, an empowered committee under the DTC, co-chaired by the chief of defence staff (CDS) and the PM's principal scientific advisor, has been recommended. The panel has also called for the bifurcation of the post of secretary, R&D, in the defence ministry. Currently, the DRDO chairman also holds the said post. These structural changes, therefore, represent the relinquishing of considerable strategic autonomy for the DRDO.

The DRDO's proposed role will now be limited to research and development, not to develop prototypes or technology demonstrators as it does currently. Tasks of production and further development are to be farmed out to selected private industry and/ or public sector undertakings. The panel has also recommended restructuring the 41 DRDO labs into 10 national ones—two each in Bengaluru and Hyderabad, and the rest in Delhi, Pune, Dehradun, Chennai, Visakhapatnam and Chandigarh.

The area where the scientists' pushback is the greatest is the importance given to private defence industry, start-ups and academic institutions. Besides the 10 national labs, the panel also proposes the setting up of five national test facilities where private players can test their newly developed weapon systems. This licence to use government assets will spare private industry the need to invest in land, machinery or other support infrastructure. The panel report has also advised the setting up of a national defence technology roadmap along with defence tech hubs in top academic institutions.



**DUD BIRD** A DRDO-made Tapas BH-201 drone. **After 13 years and nearly Rs 1,800 crore, it has failed to meet expectations**

## A SHAKE-UP OF DEFENCE R&D

- Till now, it has control over funds dedicated to defence R&D
- Little accountability for DRDO scientists thus far
- DRDO has 7,500 scientists, around 10,000 research and technical staff; 12,500 administrative, support staff. DRDO staff is hired directly
- Other than specific works farmed out to private players and tightly controlled by DRDO, there is no role for the private sector